

Solve the following system:

$$e^{2x} + 2\ln y = 6$$

$$e^{4x} + \ln y = 3$$

$$\ln y = 3 - e^{4x}$$

$$\Rightarrow e^{2x} + 2(3 - e^{4x}) = 6$$

$$e^{2x} + 6 - 2e^{4x} = 6$$

$$\text{Let } t = e^{2x}$$

$$\Rightarrow 2t^2 - t = 0$$

$$t_1 = \frac{1}{2} \text{ and } t_2 = 0$$

$$\text{D.F: } y > 0$$

$$e^{2x} = \frac{1}{2}$$

&

$$e^{2x} = 0$$

rejected

$$e > 0$$

$$\frac{1}{2} + 2\ln y = 6$$

$$2\ln y = 5.5$$

$$\ln y = \frac{11}{4}$$

$$\boxed{y = e^{\frac{11}{4}}}$$

Calculate x

$$e^{2x} = \frac{1}{2}$$

$$2x = \ln\left(\frac{1}{2}\right)$$

$$x = \frac{\ln\left(\frac{1}{2}\right)}{2}$$

$$x = \frac{\ln(1) - \ln(2)}{2}$$

$$\boxed{x = \frac{-\ln(2)}{2}}$$

$$S: \left\{ \left(\frac{-\ln(2)}{2}, e^{1/2} \right) \right\}$$